

6.3.2 Backward error analysis of dot product: general case



6.3.2 Backward error analysis of dot product

fit width



Consider now

$$\kappa := x^T y = \begin{pmatrix} \chi_0 \\ \chi_1 \\ \vdots \\ \chi_{n-2} \\ \chi_{n-1} \end{pmatrix}^T \begin{pmatrix} \psi_0 \\ \psi_1 \\ \vdots \\ \psi_{n-2} \\ \psi_{n-1} \end{pmatrix} = \left((\chi_0 \psi_0 + \chi_1 \psi_1 + \cdots) + \chi_{n-2} \psi_{n-2} \right) + \chi_{n-1} \psi_{n-1}.$$

Under the computational model given in Subsection 6.2.3 the computed result, $\tilde{\kappa}$, satisfies

$$\begin{aligned} \tilde{\kappa} &= \left(\left((\chi_0 \psi_0 (1 + \epsilon_*^{(0)}) + \chi_1 \psi_1 (1 + \epsilon_*^{(1)})) (1 + \epsilon_+^{(1)}) + \cdots \right) (1 + \epsilon_+^{(n-2)}) \right. \\ &\quad \left. + \chi_{n-1} \psi_{n-1} (1 + \epsilon_*^{(n-1)}) \right) (1 + \epsilon_+^{(n-1)}) \\ &= \chi_0 \psi_0 (1 + \epsilon_*^{(0)}) (1 + \epsilon_+^{(1)}) (1 + \epsilon_+^{(2)}) \cdots (1 + \epsilon_+^{(n-1)}) \\ &\quad + \chi_1 \psi_1 (1 + \epsilon_*^{(1)}) (1 + \epsilon_+^{(1)}) (1 + \epsilon_+^{(2)}) \cdots (1 + \epsilon_+^{(n-1)}) \\ &\quad + \chi_2 \psi_2 (1 + \epsilon_*^{(2)}) \quad (1 + \epsilon_+^{(2)}) \cdots (1 + \epsilon_+^{(n-1)}) \\ &\quad + \cdots \\ &\quad + \chi_{n-1} \psi_{n-1} (1 + \epsilon_*^{(n-1)}) \quad (1 + \epsilon_+^{(n-1)}) \\ &= \sum_{i=0}^{n-1} \left(\chi_i \psi_i (1 + \epsilon_*^{(i)}) \prod_{j=i}^{n-1} (1 + \epsilon_+^{(j)}) \right) \end{aligned}$$

so that

$$\tilde{\kappa} = \sum_{i=0}^{n-1} \left(\chi_i \psi_i (1 + \epsilon_*^{(i)}) \prod_{j=i}^{n-1} (1 + \epsilon_+^{(j)}) \right), \tag{6.3.2}$$

where $\epsilon_+^{(0)} = 0$ and $|\epsilon_*^{(0)}|, |\epsilon_*^{(j)}|, |\epsilon_+^{(j)}| \leq \epsilon_{\text{mach}}$ for $j = 1, \dots, n-1$.

Clearly, a notation to keep expressions from becoming unreadable is desirable. For this reason we introduce the symbol θ_j :

6.3.2 Backward error analysis: accumulating error factors

fit width



Lemma 6.3.2.1. Let $\epsilon_i \in \mathbb{R}$, $0 \leq i \leq n-1$, $n\epsilon_{\text{mach}} < 1$, and $|\epsilon_i| \leq \epsilon_{\text{mach}}$. Then $\exists \theta_n \in \mathbb{R}$ such that

$$\prod_{i=0}^{n-1} (1 + \epsilon_i)^{\pm 1} = 1 + \theta_n,$$

with $|\theta_n| \leq n\epsilon_{\text{mach}} / (1 - n\epsilon_{\text{mach}})$.

Here the ± 1 means that on an individual basis, the term is either used in a multiplication or a division. For example

$$(1 + \epsilon_0)^{\pm 1} (1 + \epsilon_1)^{\pm 1}$$

might stand for

$$(1 + \epsilon_0)(1 + \epsilon_1) \quad \text{or} \quad \frac{(1 + \epsilon_0)}{(1 + \epsilon_1)} \quad \text{or} \quad \frac{(1 + \epsilon_1)}{(1 + \epsilon_0)} \quad \text{or} \quad \frac{1}{(1 + \epsilon_1)(1 + \epsilon_0)}$$

so that this lemma can accommodate an analysis that involves a mixture of the Standard and Alternative Computational Models (SCM and ACM).

Proof.

By Mathematical Induction.

- Base case. $n = 1$. Trivial.
- Inductive Step. The Inductive Hypothesis (I.H.) tells us that for all $\epsilon_i \in \mathbb{R}$, $0 \leq i \leq n - 1$, $n\epsilon_{\text{mach}} < 1$, and $|\epsilon_i| \leq \epsilon_{\text{mach}}$, there exists a $\theta_n \in \mathbb{R}$ such that

$$\prod_{i=0}^{n-1} (1 + \epsilon_i)^{\pm 1} = 1 + \theta_n, \text{ with } |\theta_n| \leq n\epsilon_{\text{mach}} / (1 - n\epsilon_{\text{mach}}).$$

We will show that if $\epsilon_i \in \mathbb{R}$, $0 \leq i \leq n$, $(n + 1)\epsilon_{\text{mach}} < 1$, and $|\epsilon_i| \leq \epsilon_{\text{mach}}$, then there exists a $\theta_{n+1} \in \mathbb{R}$ such that

$$\prod_{i=0}^n (1 + \epsilon_i)^{\pm 1} = 1 + \theta_{n+1}, \text{ with } |\theta_{n+1}| \leq (n + 1)\epsilon_{\text{mach}} / (1 - (n + 1)\epsilon_{\text{mach}}).$$

- Case 1: The last term comes from the application of the SCM.

$$\prod_{i=0}^n (1 + \epsilon_i)^{\pm 1} = \prod_{i=0}^{n-1} (1 + \epsilon_i)^{\pm 1} (1 + \epsilon_n). \text{ See } \textcolor{red}{\text{Ponder This 6.3.2.1}}.$$

- Case 2: The last term comes from the application of the ACM.

$\prod_{i=0}^n (1 + \epsilon_i)^{\pm 1} = (\prod_{i=0}^{n-1} (1 + \epsilon_i)^{\pm 1}) / (1 + \epsilon_n)$. By the I.H. there exists a θ_n such that $(1 + \theta_n) = \prod_{i=0}^{n-1} (1 + \epsilon_i)^{\pm 1}$ and $|\theta_n| \leq n\epsilon_{\text{mach}} / (1 - n\epsilon_{\text{mach}})$. Then

$$\frac{\prod_{i=0}^{n-1} (1 + \epsilon_i)^{\pm 1}}{1 + \epsilon_n} = \frac{1 + \theta_n}{1 + \epsilon_n} = 1 + \underbrace{\frac{\theta_n - \epsilon_n}{1 + \epsilon_n}}_{\theta_{n+1}},$$

which tells us how to pick θ_{n+1} . Now

$$\begin{aligned} |\theta_{n+1}| &= < \text{definition of } \theta_{n+1} > \\ |(\theta_n - \epsilon_n) / (1 + \epsilon_n)| &\leq < |\theta_n - \epsilon_n| \leq |\theta_n| + |\epsilon_n| \leq |\theta_n| + \epsilon_{\text{mach}} > \\ (|\theta_n| + \epsilon_{\text{mach}}) / (1 + \epsilon_n) &\leq < |1 + \epsilon_n| \geq 1 - |\epsilon_n| \geq 1 - \epsilon_{\text{mach}} > \\ (|\theta_n| + \epsilon_{\text{mach}}) / (1 - \epsilon_{\text{mach}}) &\leq < \text{bound } |\theta_n| \text{ using I.H.} > \\ (\frac{n\epsilon_{\text{mach}}}{1 - n\epsilon_{\text{mach}}} + \epsilon_{\text{mach}}) / (1 - \epsilon_{\text{mach}}) &= < \text{algebra} > \\ (n\epsilon_{\text{mach}} + (1 - n\epsilon_{\text{mach}})\epsilon_{\text{mach}}) / ((1 - n\epsilon_{\text{mach}})(1 - \epsilon_{\text{mach}})) &= < \text{algebra} > \\ ((n + 1)\epsilon_{\text{mach}} - n\epsilon_{\text{mach}}^2) / (1 - (n + 1)\epsilon_{\text{mach}} + n\epsilon_{\text{mach}}^2) &\leq < \text{increase numerator; decrease denominator} > \\ ((n + 1)\epsilon_{\text{mach}}) / (1 - (n + 1)\epsilon_{\text{mach}}). \end{aligned}$$

- By the Principle of Mathematical Induction, the result holds.

Ponder This 6.3.2.1. Complete the proof of [Lemma 6.3.2.1](#).

Remark 6.3.2.2. The quantity θ_n will be used throughout these notes. It is not intended to be a specific number. Instead, it is an order of magnitude identified by the subscript n , which indicates the number of error factors of the form $(1 + \epsilon_i)$ and/or $(1 + \epsilon_i)^{-1}$ that are grouped together to form $(1 + \theta_n)$.

Since we will often encounter the bound on $|\theta_n|$ that appears in [Lemma 6.3.2.1](#) we assign it a symbol as follows:

Definition 6.3.2.3. For all $n \geq 1$ and $n\epsilon_{\text{mach}} < 1$, define

$$\gamma_n = n\epsilon_{\text{mach}} / (1 - n\epsilon_{\text{mach}}).$$

With this notation, [\(6.3.2\)](#) simplifies to

$$\begin{aligned}
& \tilde{\kappa} \\
&= \\
& \chi_0 \psi_0 (1 + \theta_n) + \chi_1 \psi_1 (1 + \theta_n) + \cdots + \chi_{n-1} \psi_{n-1} (1 + \theta_2) \\
&= \\
& \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \\ \vdots \\ \chi_{n-1} \end{pmatrix}^T \begin{pmatrix} (1 + \theta_n) & 0 & 0 & \cdots & 0 \\ 0 & (1 + \theta_n) & 0 & \cdots & 0 \\ 0 & 0 & (1 + \theta_{n-1}) & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & (1 + \theta_2) \end{pmatrix} \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \vdots \\ \psi_{n-1} \end{pmatrix} \\
&= \\
& \begin{pmatrix} \chi_0 \\ \chi_1 \\ \chi_2 \\ \vdots \\ \chi_{n-1} \end{pmatrix}^T \underbrace{\begin{pmatrix} \theta_n & 0 & 0 & \cdots & 0 \\ 0 & \theta_n & 0 & \cdots & 0 \\ 0 & 0 & \theta_{n-1} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \theta_2 \end{pmatrix}}_{I + \Sigma^{(n)}} \begin{pmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \vdots \\ \psi_{n-1} \end{pmatrix} \\
&= \\
& x^T (I + \Sigma^{(n)}) y,
\end{aligned} \tag{6.3.3}$$

where $|\theta_j| \leq \gamma_j, j = 2, \dots, n$.

Remark 6.3.2.4. Two instances of the symbol θ_n , appearing even in the same expression, typically do not represent the same number. For example, in [\(6.3.3\)](#) a $(1 + \theta_n)$ multiplies each of the terms $\chi_0 \psi_0$ and $\chi_1 \psi_1$, but these two instances of θ_n , as a rule, do not denote the same quantity. In particular, one should be careful when factoring out such quantities.

6.3.2 Accumulating error factors, Part 2

fit width



As part of the analyses the following bounds will be useful to bound error that accumulates:

Lemma 6.3.2.5. *If $n, b \geq 1$ then $\gamma_n \leq \gamma_{n+b}$ and $\gamma_n + \gamma_b + \gamma_n \gamma_b \leq \gamma_{n+b}$.*

This lemma will be invoked when, for example, we want to bound $|\epsilon|$ such that $1 + \epsilon = (1 + \epsilon_1)(1 + \epsilon_2) = 1 + (\epsilon_1 + \epsilon_2 + \epsilon_1 \epsilon_2)$ knowing that $|\epsilon_1| \leq \gamma_n$ and $|\epsilon_2| \leq \gamma_b$.

Homework 6.3.2.2. Prove [Lemma 6.3.2.5](#).

▼ [Solution](#)

$$\begin{aligned}
\gamma_n &= < \text{definition} > \\
(n\epsilon_{\text{mach}})/(1 - n\epsilon_{\text{mach}}) &\leq < b \geq 1 > \\
((n + b)\epsilon_{\text{mach}})/(1 - n\epsilon_{\text{mach}}) &\leq < 1/(1 - n\epsilon_{\text{mach}}) \leq 1/(1 - (n + b)\epsilon_{\text{mach}}) \text{ if } (n + b)\epsilon_{\text{mach}} < 1 > \\
((n + b)\epsilon_{\text{mach}})/(1 - (n + b)\epsilon_{\text{mach}}) &= < \text{definition} > \\
\gamma_{n+b}.
\end{aligned}$$

and

$$\begin{aligned}
&\gamma_n + \gamma_b + \gamma_n \gamma_b \\
&= \text{< definition >} \\
&\frac{n\epsilon_{\text{mach}}}{1-n\epsilon_{\text{mach}}} + \frac{b\epsilon_{\text{mach}}}{1-b\epsilon_{\text{mach}}} + \frac{n\epsilon_{\text{mach}}}{(1-n\epsilon_{\text{mach}})} \frac{b\epsilon_{\text{mach}}}{(1-b\epsilon_{\text{mach}})} \\
&= \text{< algebra >} \\
&\frac{n\epsilon_{\text{mach}}(1-b\epsilon_{\text{mach}}) + (1-n\epsilon_{\text{mach}})b\epsilon_{\text{mach}} + bn\epsilon_{\text{mach}}^2}{(1-n\epsilon_{\text{mach}})(1-b\epsilon_{\text{mach}})} \\
&= \text{< algebra >} \\
&\frac{n\epsilon_{\text{mach}} - bn\epsilon_{\text{mach}}^2 + b\epsilon_{\text{mach}} - bn\epsilon_{\text{mach}}^2 + bn\epsilon_{\text{mach}}^2}{1-(n+b)\epsilon_{\text{mach}} + bn\epsilon_{\text{mach}}^2} \\
&= \text{< algebra >} \\
&\frac{(n+b)\epsilon_{\text{mach}} - bn\epsilon_{\text{mach}}^2}{1-(n+b)\epsilon_{\text{mach}} + bn\epsilon_{\text{mach}}^2} \\
&\leq \text{< } bn\epsilon_{\text{mach}}^2 > 0 > \\
&\frac{(n+b)\epsilon_{\text{mach}}}{1-(n+b)\epsilon_{\text{mach}} + bn\epsilon_{\text{mach}}^2} \\
&\leq \text{< } bn\epsilon_{\text{mach}}^2 > 0 > \\
&\frac{(n+b)\epsilon_{\text{mach}}}{1-(n+b)\epsilon_{\text{mach}}} \\
&= \text{< definition >} \\
&\gamma_{n+b}
\end{aligned}$$